

Que-1.

Create a SQL script to insert a new order into an Orders table, then use COMMIT to save the transaction and verify that the new order persists after reconnecting to the database.

Ans:-

```
INSERT INTO Orders (  
    order_id,  
    user_id,  
    restaurant_id,  
    order_total  
)  
VALUES (  
    101,  
    1,  
    10,  
    1250.00  
);
```

COMMIT;

Que-2.

Simulate a Zomato-style food order process: insert two new items into an OrderItems table, then use ROLLBACK to undo the changes before committing. Check that no new items remain in the table after rollback.

Ans:-

```
INSERT INTO OrderItems (  
    item_id,  
    order_id,  
    item_name,  
    quantity  
)  
VALUES  
(1, 101, 'Paneer Pizza', 2);
```

```
INSERT INTO OrderItems (  
    item_id,  
    order_id,  
    item_name,  
    quantity  
)  
VALUES  
(2, 101, 'Cold Drink', 1);
```

Que-3.

In a Flipkart-like shopping cart scenario, use SAVEPOINT to mark a point after adding a product to the Cart table, add another product, then use ROLLBACK TO SAVEPOINT to undo only the last addition. Show the final contents of the Cart table.

Ans:-

Add Frist Product:-

```
INSERT INTO Cart (  
    cart_id,  
    product_name,  
    quantity  
)  
VALUES (  
    1,  
    'Wireless Mouse',  
    1  
);
```

Create a Savepoint:-

```
SAVEPOINT first_product_added;
```

Add Second Product:-

```
INSERT INTO Cart (  
    cart_id,  
    product_name,  
    quantity  
)  
VALUES (  
    2,  
    'Bluetooth Keyboard',  
    1  
);
```

Roll-Back Product:-

```
ROLLBACK TO first_product_added;
```

Show cart After Roll-Back:-

```
SELECT * FROM Cart;
```

Que-4.

Write a trigger on a Wallet table that automatically deducts the purchase amount from the user's balance whenever a new transaction is inserted, similar to how Paytm updates wallet balance after a payment.

Ans:-

```
CREATE OR REPLACE TRIGGER trg_deduct_wallet_balance
AFTER INSERT ON Transactions
FOR EACH ROW
BEGIN
    UPDATE Wallet
    SET balance = balance - :NEW.purchase_amount
    WHERE user_id = :NEW.user_id;
END;
/
```

Insert New Transaction:-

```
INSERT INTO Transactions
(transaction_id, user_id, purchase_amount)
VALUES
(1, 101, 1200);
```

Que-5.

Use ChatGPT to generate a SQL test scenario where a trigger incorrectly updates a user's balance after an expense entry. Paste the scenario and your corrected trigger code, explaining how you fixed the error.

Hint: Ask ChatGPT for a buggy trigger example and a test case that exposes the bug.

Ans:-

Incorrected Trigger Give chatGPT:-

```
CREATE OR REPLACE TRIGGER trg_update_balance
AFTER INSERT ON Expenses
FOR EACH ROW
BEGIN
    UPDATE Wallet
    SET balance = balance + :NEW.amount
    WHERE user_id = :NEW.user_id;
END;
/
```

Trigger Test

```
INSERT INTO Expenses
VALUES (1, 101, 1000);
```

Solved Trigger:-

```
CREATE OR REPLACE TRIGGER trg_update_balance
AFTER INSERT ON Expenses
FOR EACH ROW
BEGIN
    UPDATE Wallet
    SET balance = balance - :NEW.amount
    WHERE user_id = :NEW.user_id;
END;
/
```